Rapid Surface Deployment of a DAS System for Earthquake Hazard Assessment

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Abstract: Distributed acoustic sensing (DAS) is a relatively new technology used in many geophysical applications. The versatility, high spatiotemporal resolution, and sensitivity to surface waves make DAS ideal for rapid deployment surveys such as earthquake-aftershock monitoring and hazard assessment. However, these applications usually rely on trenched cable installations that are time consuming and cost-prohibitive to deploy, or on existing telecommunication fibers that are limited in spatial coverage. To examine the potential of untrenched surface deployments, we acquired DAS data in continuous mode for one hour on a rapidly deployed fiber array composed of six parallel linear subsections directly on the surface with different fiber-ground contact conditions. We applied ambient interferometry and adopted a simplified method to determine the average shear-wave velocity of the top 30 m (V_{S30}). Our methodology resulted in robust V_{S30} estimates for each surface deployment subsection that are consistent with collocated 1-m-deep trenched cables. The implications of these findings support DAS as a viable method for noninvasive-deployment surface surveys for rapid earthquake hazard and damage assessment. **DOI: 10.1061/JGGEFK.GTENG-10896.** *© 2023 American Society of Civil Engineers.*

Introduction

Variations in near-surface soft soil deposits are thought to largely dictate seismic amplitudes on a local scale (Kuo et al. 2012; Najaftomarei et al. 2020; Zhan 2020). The time-averaged shear-wave velocity of the top 30 m (V_{S30}) has been the quantitative parameter adopted by the National Earthquake Hazards Reduction Program (NEHRP) for site classification and building codes in the United States for nearly two decades (Brown et al. 2000; Comina et al. 2011; Yong et al. 2016). Site class is used to characterize soil properties and determine the local seismic coefficients for earthquake-resistant structural design (Dean and Seaton 2005; Comina et al. 2011). For large-scale commercial buildings and city infrastructure, the difference between site classifications can substantially affect the initial and long-term cost of construction (Dean and Seaton 2005). The V_{S30} index has also been widely used as a parameter for characterizing effects of soil stiffness on ground-motion prediction studies (Abrahamson and Silva 1997; Power et al. 2008; Boore et al. 2011).

Traditional V_{S30} surveys generally are invasive, use logistically challenging borehole measurements (i.e., crosshole, suspension logging methods), and require prior authorization to implement (Comina et al. 2011; Garofalo et al. 2016; Yong et al. 2016). Accordingly, current V_{S30} maps are spatially limited in coverage. In areas where no V_{S30} data are available, estimates often rely on interpolated values or broadband and long-period permanent seismic arrays that suffer from reliability and resolution issues due to poor station distribution (Zhan 2020). The development of noninvasive techniques that measure surface-wave dispersion properties address many of the challenges associated with V_{S30} surveys and greatly improve data coverage and availability for structural engineers. Within the last two decades, methods such as the spectral analysis of surface waves (SASW) (Dean and Seaton 2005; Dai and Li 2014; Garofalo et al. 2016; Lin et al. 2021), and statistical methods (Brown et al. 2000) have been widely used for cost-effective, noninvasive site classification studies.

In recent years, distributed acoustic sensing (DAS) has emerged as a reliable tool for near-surface applications and subsurface characterization. Its versatility and high spatial-temporal resolution make it a suitable instrument for noninvasive V_{S30} surveys. DAS uses the principles of time-domain reflectometry to effectively turn a length of a fiber-optic cable into a linear network of seismic sensors out to tens of kilometers (Zhan 2020). Dynamic strain is recorded along the fiber by measuring the phase difference of backscattered light caused by impurities in the fiber core. DAS is mostly sensitive to particle motion along the axis of the fiber, making it highly effective for measuring the horizontal component of surface waves (i.e., Rayleigh). Recent applications use preexisting unused telecommunication networks (i.e., so-called dark fiber) to record high-quality seismic data from earthquakes (Lindsey et al. 2019; Ajo-Franklin et al. 2019; Li et al. 2021) as well as ambient and anthropogenic sources in dense urban areas (Shragge et al. 2021; Winters et al. 2021; Zhu et al. 2021). In locations where dark fiber is unavailable, traditional deployment consisting of burying (<1 m) DAS cables (Dou et al. 2017; Luo et al. 2020) is used to ensure sufficient coupling and higher signal-to-noise ratios. However, this approach often proves cost-prohibitive and logistically challenging to implement.

Research concerning the instrument response of DAS cables deployed directly on the ground surface is sparse. Spikes et al. (2019) reported that helically wrapped cables draped along the ground surface recorded active-source (i.e., hammer-shot) reflection data of similar quality to data recorded by collocated geophones.

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High-fidelity data acquisition from surface DAS arrays could provide geotechnical engineers with an efficient and cost-effective method for numerous near-surface applications. In this study, we compared ambient waveform data recorded by a surface-deployed DAS array with data recorded from collocated trenched cables. We applied a simplified approach to determine V_{S30} directly from the dispersion spectra for linear subsections of both arrays, which avoided the computationally expensive inversion analysis inherent in traditional methods.

The paper begins with a description of the unique deployment of the DAS arrays and the data acquisition parameters used for the field study. We then discuss our data-processing approach, which includes ambient-data interferometry, dispersion curve mapping, and generation of V_{S30} estimates. This is followed by a presentation and discussion of our results for each data-processing step. We find that further analysis of the ambient waveforms and dispersion curves provides important qualitative information such as signal coherency and dispersion features to compare coupling conditions. A quantitative comparison of our V_{S30} results shows that surface-deployed DAS cables are viable for near-surface geotechnical surveys. We conclude with potential implications of surface DAS fiber deployment.

Field Description and Data Acquisition

The field test was conducted at Kafadar Commons located on the campus of the Colorado School of Mines. Kafadar Commons is a rectangular grass field approximately 40×100 m in size. At the time of data acquisition, the southern half of the field was covered by several inches of snow while the northern side was exposed grass.

We deployed 650 m of optical fiber along the surface of the field and connected it to the 1-km trenched DAS array previously installed beneath Kafadar Commons as part of the Mines Underground Geophysical Laboratory (see Fig. 1) (Krahenbuhl et al. 2018). The surface array was deployed as six parallel lines roughly 100 m in length using two coupling methods: (1) laid directly on the ground; and (2) pressed into the ground by several people walking on top of the cable. Cables deployed under Condition 2 are referred to here as "pressed." The first line segment was draped along the concrete walking path north of the field (Concrete North), followed by two lines along the grass (Grass Pressed and Grass), and two lines along the snow (Snow and Snow Pressed). The final line segment was deployed along the concrete walking path south of the field (Concrete South) and terminates inside the adjacent CoorsTek building [Fig. 1(a)]. Parallel line segments were separated by approximately 2.5 m, except the Concrete North [Fig. 1(b)] and Grass Pressed [Fig. 1(c)] segments which were separated by approximately 7.5 m, and the Grass [Fig. 1(c)] and Snow Pressed segments separated by approximately by 30 m [Fig. 1(d)]. The overall footprint of the surface array was 45×100 m.

The trenched array followed a 27×90-m grid pattern located 1 m beneath the surface (Luo et al. 2020). To examine nearly coincident instrument responses, we used only data recorded along the four subsections (Subsurfaces 1–4) running parallel to the surface array. The two subarrays were connected by splicing the end of the surface cable to the fiber-optic lead of the subsurface cable located in the CoorsTek building. We calibrated the DAS channel locations using GPS coordinates, tap tests, and hammer-shot data.

Recording data from a single continuous array allowed us to use one DAS interrogator to simultaneously monitor the entire array, which avoided any uncertainty in the acquisition parameters and data processing introduced by the use of different interrogator units. Additionally, the unique deployment allowed us to record data

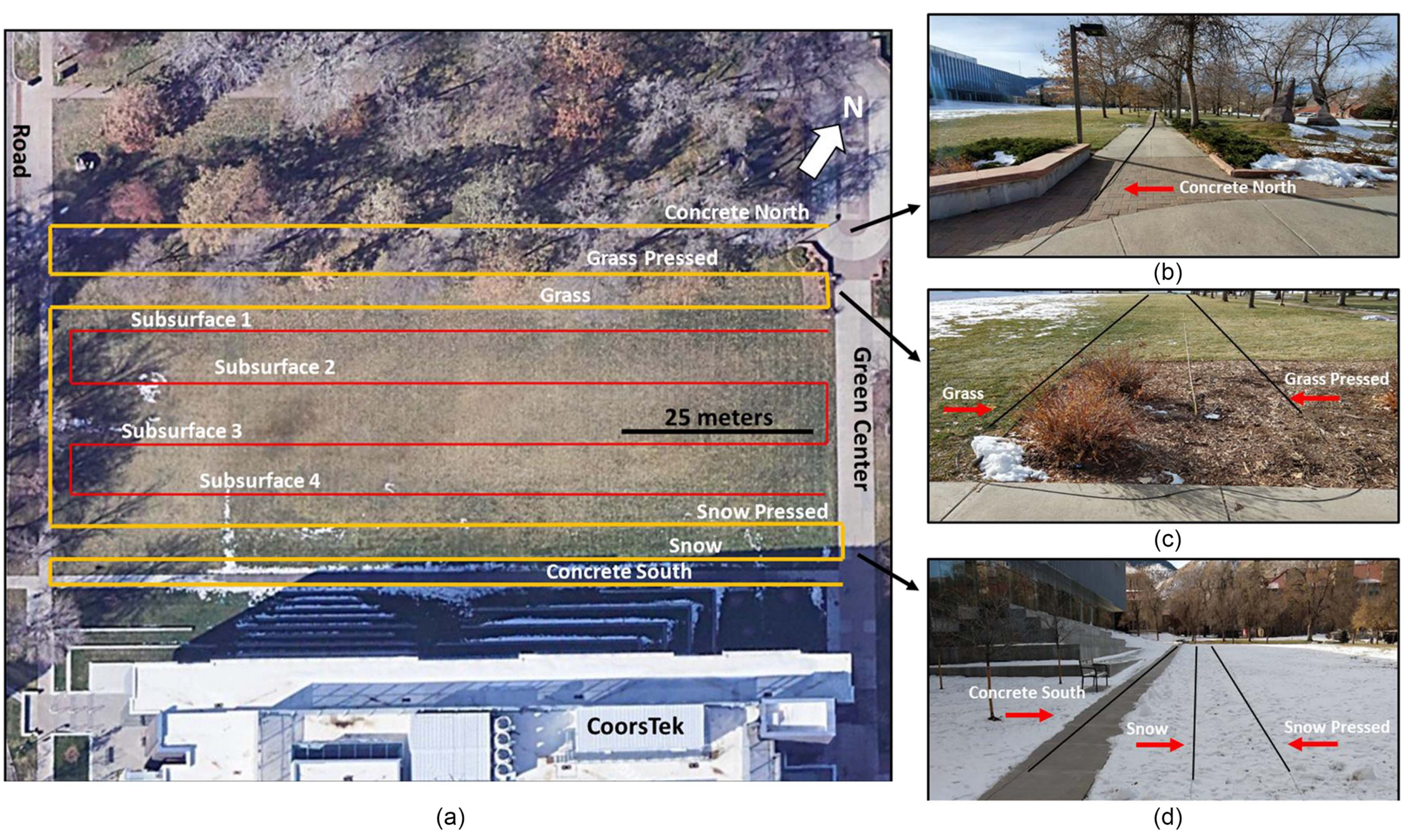

Fig. 1. DAS deployment overview: (a) map view of Kafadar Commons showing DAS surface and subsurface deployments; (b) Concrete North; (c) Grass and Grass Pressed; and (d) Concrete South, Snow, and Snow Pressed. (Black lines indicate cables.)

within a relatively small footprint from subsections of fiber under different mechanical coupling conditions. In general, we anticipated an increased signal-to-noise ratio from subsections with enhanced coupling (i.e., pressed and trenched) compared to subsections laid on the ground. Although pressing the fiber into the ground provided minimal embedment (less than several centimeters), we still expected greater signal coherency compared with the adjacent subsections laid on the ground coupled by their own weight. Given the durability of the fiber optic cable, there was little risk associated with damaging the fiber while pressing it into the ground. If any damage were caused, it would be easily identified during data acquisition.

Approximately 1.0 h of data was recorded on December 2, 2020, using a Terra15 Treble interrogator unit (Terra15, Perth, Australia) at a 4,467-Hz sampling rate with a channel spacing of approximately 2.45 m. A gauge length of 4.0 times the channel spacing was used which acted as a differential operator across the spatial axis of the data to convert the measurements to strain rate equivalent. We recorded ambient waveforms from anthropogenic sources in the survey area, including a nearby road running orthogonal to our array setup approximately 5 m from the nearest DAS channel. Notably, that the entire acquisition, from deployment to teardown, took just over 3 h to complete. A crew of four inexperienced students took approximately 20 min to unspool and deploy the 650 m of fiber-optic cable without using specialized equipment. One hour was spent splicing the surface cable to the trenched cable, which is unlikely to be required in most field applications.

Methodology

Ambient-Noise Interferometry

Recent studies have demonstrated the potential of ambient-noise interferometry to characterize the near surface using DAS deployment (Lindsey et al. 2019; Shragge et al. 2021; Luo et al. 2020). Cross-correlation of diffuse wavefields (e.g., ambient-noise) can be used to effectively recover coherent seismic information, also known as empirical Green's functions. The resulting noise correlation functions describe the signal response at seismic receiver pairs, in which spectral methods can be used to determine phase velocity-frequency dependence (Bensen et al. 2007; Ajo-Franklin et al. 2019; Luo et al. 2020). With this approach, we produced virtual shot gathers for each linear profile of the surface and subsurface arrays. Each DAS channel along a linear segment of fiber was considered a virtual seismic source (U_i) or receiver (V_i) (Fig. 2). We partitioned the 1-h record of data into 2.0-s time windows and then calculated and stacked the cross-spectrum (Jin et al. 2015) according to

$$\rho(x_s, x_r, \omega) = \sum_{i=1}^{N} \frac{U_i(x_s, \omega) V_i^*(x_r, \omega)}{|U_i(x_s, \omega)||V_i(x_r, \omega)| + \epsilon} \tag{1}$$

where ω = angular frequency; U_i and V_i = Fourier transform of the ith time segment of two virtual stations; x_s and x_r = spatial locations of the virtual source and receivers, respectively; * = complex conjugate; and real positive constant ϵ = stabilization term. Individual cross-spectra were calculated for all possible station pairs, resulting in 7,162 virtual shot-receiver pairs. The 2.0-s windows were stacked in the frequency domain, with the resulting cross-spectrum returned to the time domain by applying an inverse Fourier transform. The resulting waveform had positive and negative time lags representing waves propagating in opposite directions between the virtual stations (Bensen et al. 2007). Applying this method within each linear subsection and treating each DAS channel as a virtual source produced corresponding three-dimensional (3D) waveform volumes that were analyzed for data quality prior to dispersion analysis.

Dispersion Analysis

To produce frequency–velocity dispersion images, we used the phase-shift method (Park et al. 1998). With this method, we could

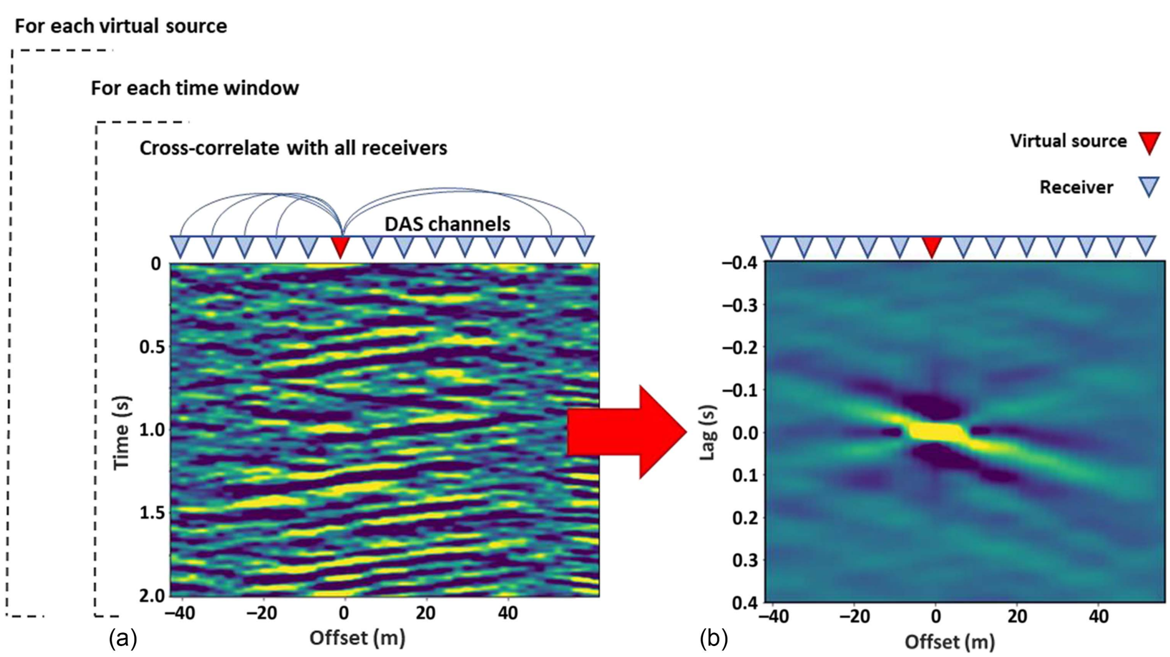

Fig. 2. Ambient-noise stacking: (a) an example of raw DAS data recorded at station offsets via 2.0-s time-domain slice; and (b) stacked noise correlation for single virtual source and receiver offsets. (Time-lag axis reduced for visibility.)

separate the different surface-wave modes despite the relatively limited number of traces and offsets from our virtual shot gathers. The first step was to extract the wavefield phase by applying a temporal Fourier transform to the virtual shot gathers (ρ) produced from Eq. (1). Next, we applied the offset-dependent phase shift and normalized the amplitudes to ensure equal weighting for each individual trace

$$P(\omega, c) = \sum_{x_s} \sum_{h} \frac{\rho(x_s, h, \omega)}{|\rho(x_s, h, \omega)|} \mathrm{e}^{-i\omega h/c} \tag{2}$$

where $h = x_s - x_r$ = vector of station offsets (Fig. 2). The maximum amplitude term occurs when signals travel at the same phase velocity (c) for a given angular frequency. This produces a frequency-velocity dispersion image for each channel serving as a virtual source along each linear profile. Because the primary goal was to produce V_{S30} estimates for each linear subsection, we stacked the dispersion images of all virtual sources from the same linear profile to produce a single dispersion image for each subsection.

V_{S30} Estimation

We estimated V_{S30} for each subsection using a simplified statistical method (Brown et al. 2000). Compared with traditional SASW, this method is computationally efficient but provides only a single V_{S30} value without using traditional inversion methods that produce one-dimensional (1D) shear-wave velocity profiles for a range of depths. For detailed information on the adopted method, including case studies and uncertainty analysis, the reader is referred to the literature (Brown et al. 2000; Lin et al. 2021). The correlation between Rayleigh-wave phase velocity and V_{S30} can be described by the predictive empirical equation (Brown et al. 2000) as

$$V_{S30} = 1.076 \times V_{R36} \tag{3}$$

where V_{R36} = Rayleigh-wave phase-velocity at a wavelength of $\lambda = 36$ m. Extracting V_{R36} from the processed dispersion curves results in a phase velocity and amplitude for discrete frequencies within the chosen frequency band. Frequency resolution is determined by two parameters: (1) the raw data sampling rate; and (2) as a consequence of the Fourier transform, the window length of the processed ambient time series. A cubic interpolation method was applied to each dispersion image to resample to a denser frequency axis prior to V_{S30} processing. We determined the final V_{S30} value for each subsection using the V_{R36} value with the highest amplitude in the dispersion image.

Results

Fig. 3 displays examples of ambient waveforms for surface array subsections [Figs. 3(a–f)] and two subsurface array subsections [Figs. 3(g and h)] after transforming the cross-spectrum to the time-space domain. Offsets for each subsection vary slightly due to differences in cable length. The observed waveform direction reversal is due to subsections being deployed in an east–west or west–east orientation. The high-amplitude zero offsets of the waveforms represent the source channel autocorrelation of each virtual shot gather. After manually scanning for data quality, we find that most virtual shot gathers exhibit coherent waveforms. Fig. 3 highlights offsets with poor signal response. Notably, the Snow subsection lacks coherent waveform features along 40 m of the fiber and, to a lesser extent, the Grass Pressed and Snow Pressed subsections display poor data quality along 12 m of fiber. The waveforms for all subsections display an asymmetric distribution, which is discussed in detail subsequently.

Fig. 4 shows the stacked, normalized dispersion curve results for the surface [Figs. 4(a–f)] and subsurface [Figs. 4(g and h)] profiles for the 3–20-Hz frequency band and a 100–2,000-m/s phase velocity range. The most stable results are within the lower frequency range of 3–7 Hz defined by the high spectral amplitudes. All dispersion curves show an abrupt discontinuity between 7 and 9 Hz, potentially indicating a mode transition commonly attributed to depth variations in soil stiffness (Foti et al. 2018). This dispersion feature was consistently observed in a recent study conducted with buried sensors in the survey area by Luo et al. (2020). Coherency in the dispersion curves begins to diminish at higher frequencies, with no continuous dispersion features above 20 Hz. The subsurface subsections show the most consistent results, with relatively well-defined spectral amplitudes for frequencies below 15 Hz. These are followed by the Concrete North, Grass, and Grass Pressed subsections that show stable results for frequencies less than 12.5 Hz. The Snow Pressed subsection is stable for frequencies less than 11 Hz, and finally the Snow and Concrete South subsections are stable for frequencies below 10 Hz.

Values of V_{R36} were estimated using the dispersion curves for each subsection. Fig. 5 shows the results for the Concrete North profile as an example. The dashed line in Fig. 5(a) represents the phase-velocity and frequency values corresponding to a wavelength of 36 m (V_{R36}). Fig. 5(b) tracks the normalized spectral amplitude as a function of phase velocity for discrete frequencies along the V_{R36} line. The marker denotes the peak amplitude which is the value used to calculate V_{S30} for each subsection (Fig. 6). Note that Grass Pressed and Concrete South returned peak spectral values along the V_{R36} line corresponding to phase velocities of 279 and 705 m/s, respectively. Prior to calculating the V_{S30} value for these subsections, we analyzed their dispersion curves and found that the second highest spectral values compared well (<3% difference) with the phase velocities of the three nearest respective subsections, the values for which are reported in Fig. 6.

We compare our results to a previous study that used more sophisticated methods to analyze ambient data acquired only from the subsurface array. Luo et al. (2020) presented a detailed 1D S-wave velocity profile using multimodal Monte Carlo inversion for the first 120 m of the subsurface. We find that their numerical results suggest a V_{S30} estimate of approximately 370 m/s for the entire survey area using all components of the subsurface array. This result falls within approximately 5% of our V_{S30} estimates along the north side of the field and within approximately 9% toward the south side (Fig. 6).

Discussion

Fig. 3 shows that the time-domain waveforms for all subsections exhibit asymmetric behavior. The observed effect is an amplitude difference between signals traveling along the positive (causal) and negative (acausal) time lags. Ideally, these signals would be identical if the sources of ambient energy were to be distributed homogeneously throughout the survey area (Bensen et al. 2007). However, the waveform asymmetry is likely due to the heavy traffic on the road directly to the west (Fig. 1). We make this interpretation based on (1) the road running perpendicular to the DAS subsections; and (2) the proximity (approximately 5 m) to the nearest DAS channel. The DAS instrument response is highly sensitive to propagating waves with particle motion oriented along the fiber axis. Thus, the primary source of energy is likely from the high-amplitude surface waves generated by nearby traffic noise.

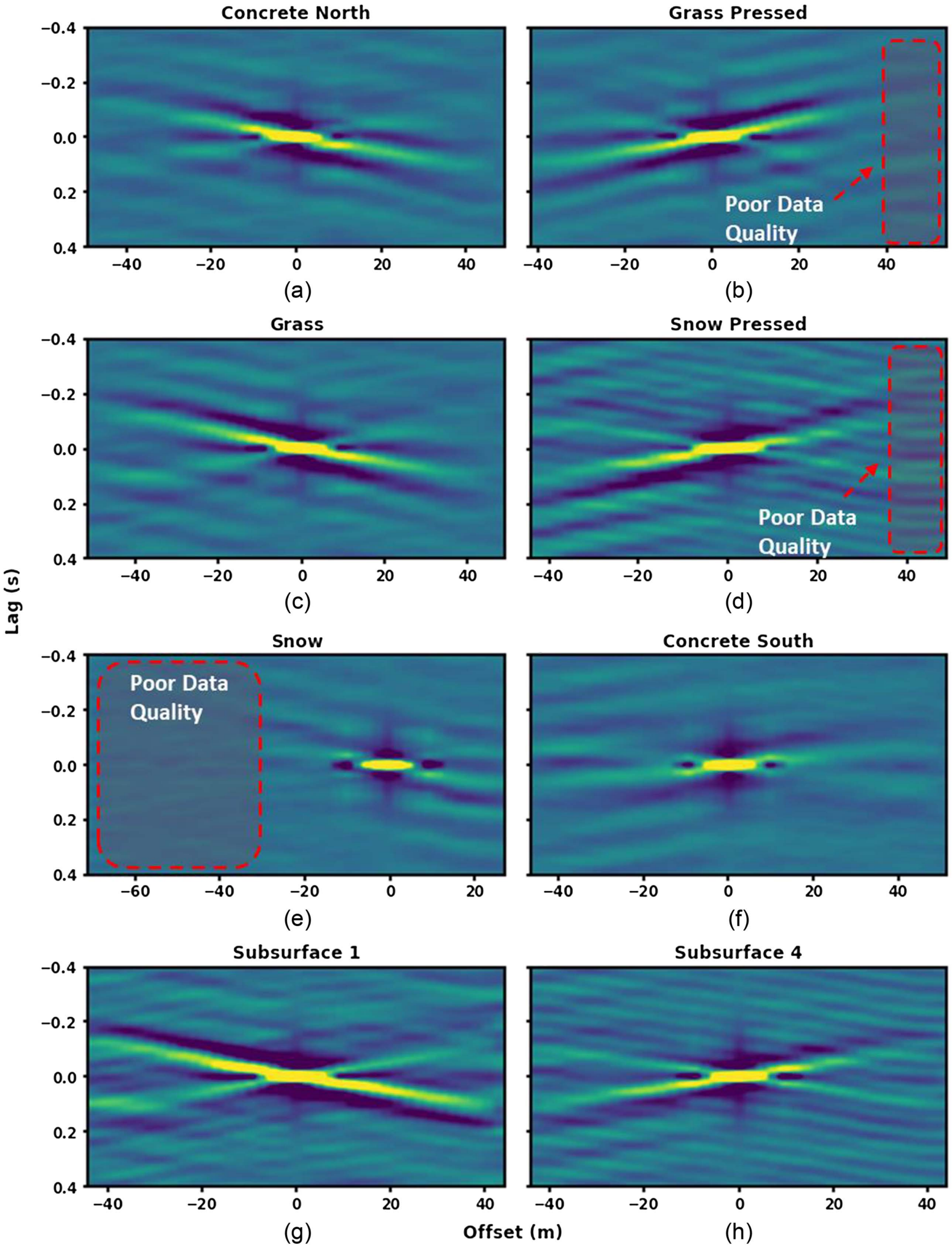

Fig. 3. Stacked correlated ambient waveforms for single virtual source in time-space domain for Surface subsections: (a) Concrete North; (b) Grass Pressed; (c) Grass; (d) Snow Pressed; (e) Snow; and (f) Concrete South; and for Subsurface subsections; (g) 1; and (h) 4. (Due to similar results, Subsurfaces 2 and 3 omitted; 2–20 Hz bandpass filter applied for visibility.)

The spectral differences presented in our results commonly arise when applying an interferometric approach to field data (Bensen et al. 2007). The asymmetry in ambient waveforms is a consequence of wavefield energy distribution and array geometry. The most effective seismic array geometries for near-surface surveys require consideration of the spatial distribution of ambient energy sources. Dominant energy sources (e.g., roads) should be of particular importance (Dou et al. 2017). This is especially true in terms of time and cost if sensors are to be trenched prior to data acquisition. The novelty of our surface deployment is that it minimizes the economic risk involved in both static and time-lapse surveys, allowing for flexible installation and rapid reconfiguration of seismic arrays if necessary.

Our study indicates that robust V_{S30} estimates can be made from linear DAS arrays deployed directly on the ground surface. Further analysis of the ambient waveforms and dispersion curves provides qualitative information describing the coupling of each subsection. In general, we observe that waveform coherency, high-amplitude moveouts, and continuous dispersion curves result from superior coupling conditions. Clear examples are the substantial difference in signal response for the Snow and Snow Pressed subsections. The response along the Snow returns a relatively low-amplitude coherent waveform with moveout at −30 to 30-m offsets along the fiber, whereas the Snow Pressed subsection results in a higher-amplitude coherent waveform with moveout across most of the fiber (90 m). We also see more continuous dispersion features for the Snow

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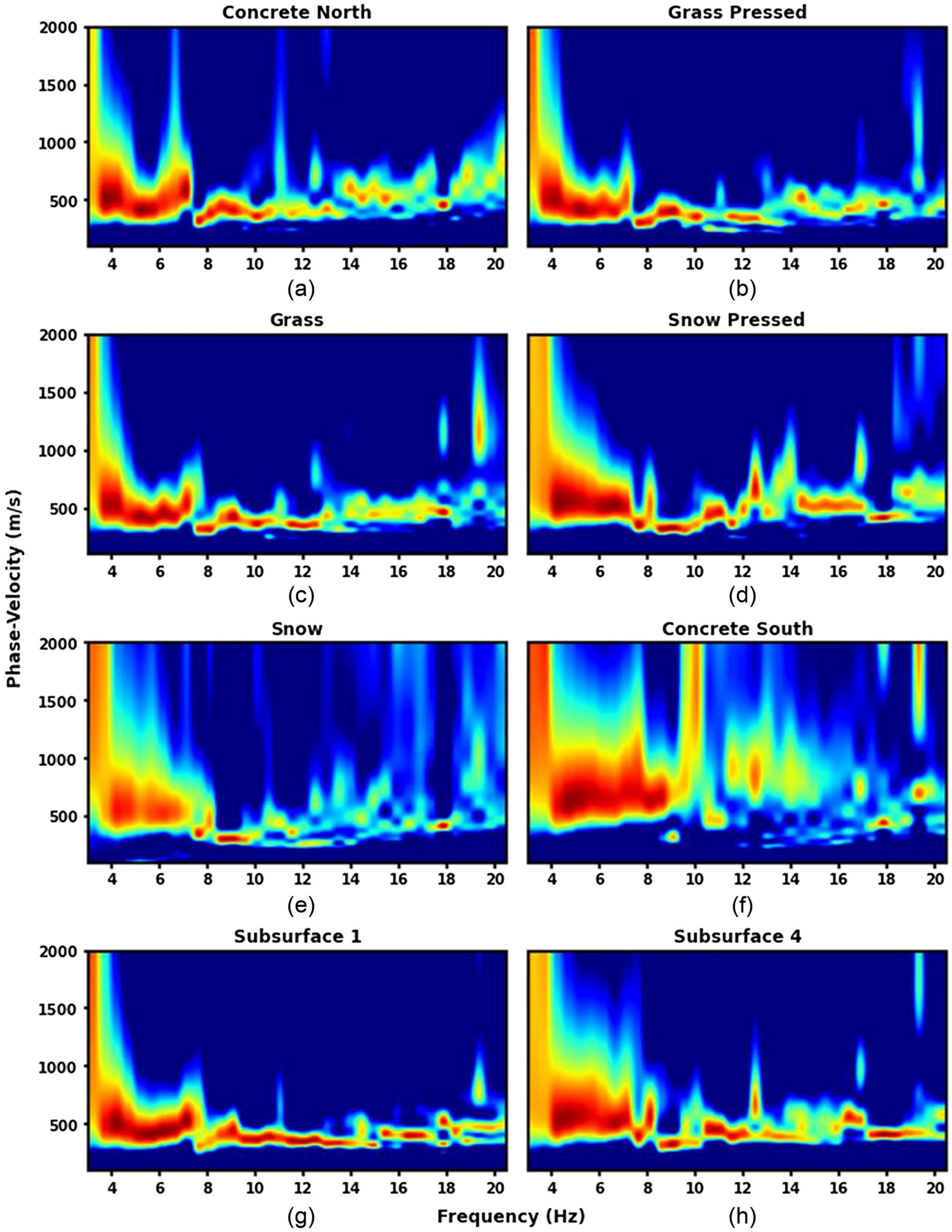

Fig. 4. Normalized and stacked dispersion spectra for Surface subsections: (a) Concrete North; (b) Grass Pressed; (c) Grass; (d) Snow Pressed; (e) Snow; and (f) Concrete South; and for Subsurface subsections: (g) 1; and (h) 4.

Pressed subsection in comparison with the Snow subsection (Fig. 4). We attribute the differences in signal response to spatial coupling variations along the Snow subsection. It is likely that local sections of the fiber are not in direct contact with the snow, limiting the number of DAS channels to record surface waves and lowering the signal-to-noise ratio. This suggests that simply pressing the fiber into the snow by walking on it improves the fiber-ground contact and significantly enhances the energy transfer and signal response. This point is demonstrated by Walter et al. (2020), who conducted a DAS survey on glaciated terrain and reported that sections of fiber closer to seismic sources resulted in a higher signal-to-noise ratio. We see a similar result in our study for offsets closer to the road for the Snow subsection.

Fig. 6 shows the V_{S30} trend with decreasing velocity from the northern to the southern halves of the field defined by a discontinuity between Subsections 1 and 2. While a full geologic interpretation is beyond the scope of this paper, a straightforward explanation for the decreasing velocity would be lateral subsurface variations. It is possible that this observation is due to differential soil compaction of the top 2 m in the field, with less consolidated soil to the south of Subsection 1 due to the Mines Underground Geophysical Laboratory installation (Krahenbuhl et al. 2018). The excavation perimeter is approximately collocated with the change in estimated V_{S30}. The primary takeaway from our study is the consistent V_{S30} estimates for collocated subsections of the surface and subsurface arrays given the different coupling conditions. We observe that estimates for the northern part of the field (Concrete North—Subsurface 1) and the southern part of the field (Subsurface 2—Concrete South) are all within 3% despite different coupling and ground conditions.

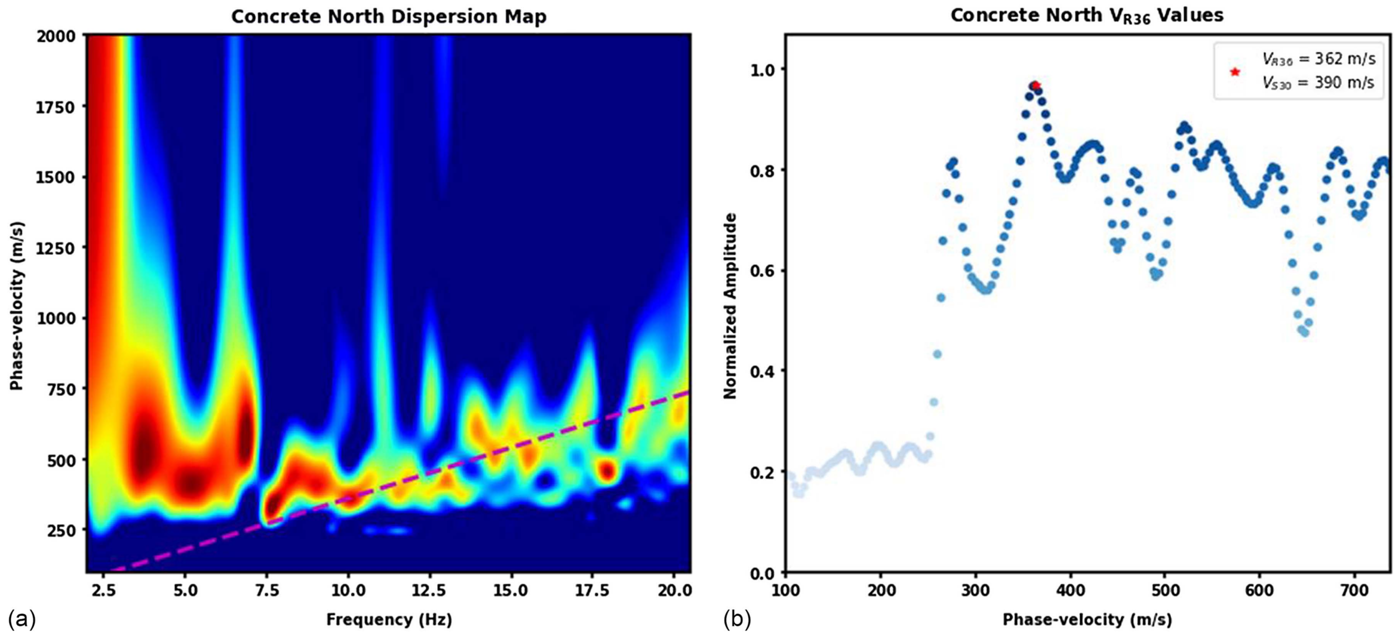

Fig. 5. Estimated V_{R36} results for Concrete North: (a) stacked normalized dispersion curve (dashed line indicates $\lambda = 36$ m for corresponding phase velocities and frequencies); and (b) spectral amplitudes for frequency values along dashed line from (a) with marker indicating peak $\lambda = 36$-m amplitude.

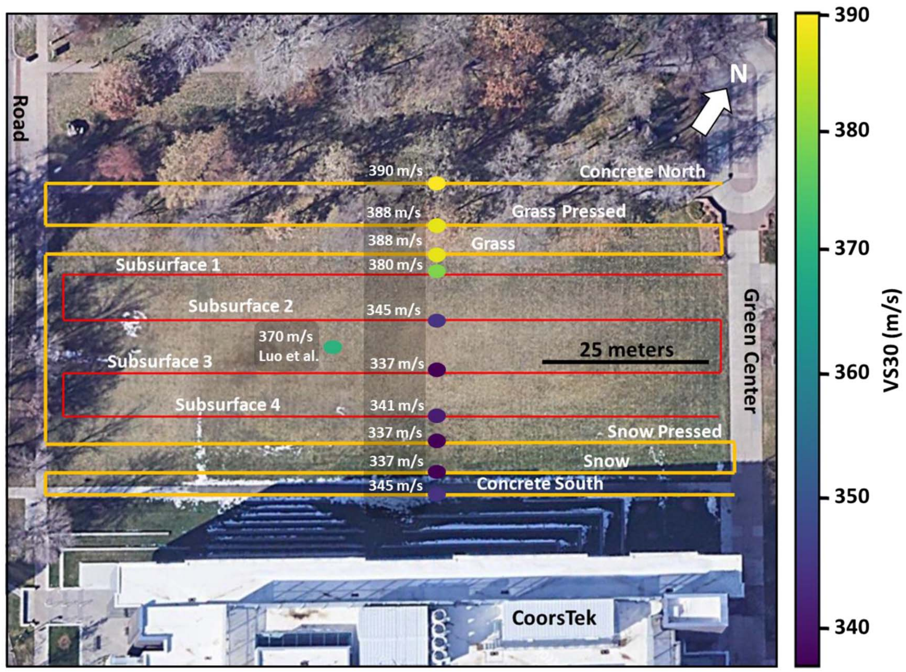

Fig. 6. V_{S30} results for each linear subsection showing lateral velocity variations across field. [Markers correspond to subsection in which V_{S30} estimates produced, not exact location. Luo et al. (2020) results estimated using all subsurface array components.]

Practical applications of our findings include several near-surface geophysical surveys. Site classification for earthquake hazard assessment can incorporate our approach in areas where there is insufficient V_{S30} coverage and where traditional seismic analysis methods are impractical and costly (i.e., urban areas). DAS surface deployments can be easily implemented by a small team in a time frame of hours as opposed to the days required by traditional trenched techniques, and they do not require access to subsurface conduits or trenching. Additionally, our methodology and the flexibility of DAS offers geotechnicians the spatial resolution to constrain lateral variations in the velocity structure throughout a survey site, as seen in our results.

Factors such as environmental impact, survey site associated risks, and acquisition parameters can limit the practicality of different deployment methods. Planting ground-penetrating instruments such as geophones with spikes may not be permitted in environmentally protected areas, such as capped landfills, where fully noninvasive surface techniques can be adopted to mitigate any potential risk to the engineered capped ground structure. Contaminated sites (e.g., former munitions facilities) often require approval as well as compliance training on contaminated soil risks prior to disturbing the ground. DAS surface deployments offer value in reducing both the lead-up time to surveys and exposure risk. Additionally, rapid deployment of dense seismic networks is crucial for acquiring critical data immediately following mainshock earthquakes (Li et al. 2021). Although modern geophone nodes are relatively easy to deploy, considerably more effort is required to match the array aperture with the spatial sampling density (1–5 m) inherent with DAS. Current DAS aftershock monitoring deployments use preexisting teleseismic networks (dark fiber), which are restrictive in availability and spatial distribution (Ajo-Franklin et al. 2019; Li et al. 2021). Shear-wave velocity characterization using permanently installed strong-motion stations also suffer the same limitations (Wong et al. 2011; Harinarayan and Kumar 2017). The ability to rapidly deploy DAS cables along the surface addresses some of these issues, providing geotechnicians with an additional data acquisition method.

Traditional surface-wave methods often rely on sophisticated inversion techniques to estimate subsurface structures. The simplified approach adopted in our study is an easily automated workflow and can be used to reduce computational expense for rapid V_{S30} estimations, providing engineers and decision makers first-order information on lateral variability of soil conditions prior to generating full V_S profiles. Finally, our deployment method is scalable to much longer distances and not restricted to the local site characterization presented in this study. Surface deployments via the towing of DAS cable along the ground (e.g., land-streamer configurations) enable kilometer-scale acquisitions, addressing some of the practical challenges of dark-fiber arrays, such as fixed deployment geometry and the direction of ambient/passive wavefield energy.

Conclusions

We demonstrated the effectiveness of DAS to record dynamic strain from surface waves generated by anthropogenic activity when deployed directly on the ground surface under different coupling conditions. We deployed a DAS array composed of six parallel linear subsections along the ground surface above preinstalled collocated trenched cables buried 1 m below. By applying an interferometric approach to 1.0 h of recorded data, we showed that the resultant waveforms and dispersion curves contain qualitative information describing the variable coupling of each linear subsection. Coherent high-amplitude waveforms, moveouts across channels, and continuous dispersion features characterize the coupling quality. Furthermore, ambient waveform asymmetry suggests that the data are influenced by the spatial distribution of dominant energy sources, highlighting the importance of flexible seismic array installations.

Adopting a simplified estimation method that does not require inversion, we produced robust V_{S30} estimates consistent for collocated subsections despite the differences in coupling conditions. Implications of this study suggest that DAS can be rapidly deployed along the ground surface for acquisition of high-quality data for a variety of near-surface seismic applications.

Data Availability Statement

Some or all data, models, or code generated or used during the study are available in a repository or online in accordance with funder data retention policies: https://doi.org/10.5061/dryad.3j9kd51k9. Table S1 is available on Dryad at https://doi.org/10.5281/zenodo.7255793.

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